

Slot 9 · Feature bake-off, model frozen

File: `features.py` · **Done when:** `uv run pytest tests/test_slot09.py` shows 5 passed · **Time:** 12 min

Today's idea

The model is **frozen**. Nobody changes it. You can only change the **columns** you give it. Which new columns actually help? The answer surprises most people.

The frozen model (do not touch):

```
lgb.LGBMRegressor(n_estimators=250, learning_rate=0.07, num_leaves=40, random_state=2026)
```

Step 1 · See the starting score

```
uv run python features.py
```

```
base columns, as shipped      MAE 5.606
```

```
now write build_features() and add your columns to FEATURES.
```

Step 2 · Add features, round by round

Write `build_features(train, other)`. It returns both tables with your new columns added. Then add the new column names to the `FEATURES` list at the bottom. Try in this order, and run after each round:

```
uv run python features.py
```

round	idea	example
1	flags	<code>is_night</code> , <code>is_peak</code> (12, 13, 19, 20), <code>is_weekend</code>
2	ratios	distance per prep minute, price per item
3	history	each restaurant's average <code>actual_minutes</code> in the past
4	more history	each courier's average; restaurant × peak-hour average

A flag looks like this:

```
tr = train.with_columns(pl.col("hour").is_in([12, 13, 19, 20]).alias("is_peak"))
```

A history column looks like this — **learn it from `train` only**, then join it on **both**:

```
rest = train.group_by("restaurant_id").agg(
    pl.col("actual_minutes").mean().alias("rest_hist_mean"))
tr = train.join(rest, on="restaurant_id", how="left")
ot = other.join(rest, on="restaurant_id", how="left")
# a restaurant never seen in train gets null -> fill it with the train average
```

Careful Never compute an average over `other` (or over `train + other` together). That is cheating: the model gets to peek at the answers. The test `test_fitted_on_train_only` checks this.

Step 3 · The leaderboard

Tell the instructor your best MAE and your best feature. Reference results:

base columns	5.606	
+ flags	5.589	(gain 0.017)
+ ratios	5.589	(gain 0.000)
+ history (train only)	5.126	(gain 0.463)
+ restaurant x peak history	5.094	(gain 0.032)

Think Flags and ratios gave almost **nothing**. History gave **everything**. Why? (Hint: does a flag give the model any information it did not already have?)

Step 4 · Check you are done

```
uv run pytest tests/test_slot09.py
```

If something goes wrong

what you see	what to do
MAE below about 4.5	Suspicious! You probably used a column that is only known after delivery. Slot 10 is about this.
<code>test_fitted_on_train_only</code> fails	You aggregated over <code>other</code> . Aggregate over <code>train</code> only, join onto both.
nulls after the join	Restaurant or courier not in train. <code>fill_null(train["actual_minutes"].mean())</code>
<code>KeyError</code> / column not found	A name in <code>FEATURES</code> is not a column your <code>build_features</code> made.